Ronquoz district energy report
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Executive summary
- Buildings analysed: 196
- Total SRE: 744 731 m2
- Annual heat + ECS: 26 147.8 MWh/an
- Peak heat demand (hourly): 9 259 kW
- Diversified peak (foisonnement): 7 222 kW
- Annual district electricity load: 23 371.9 MWh/an
- Annual PV production: 19 318.9 MWh/an
- Network length: 14.77 km
- Network CAPEX: 16 736 196 CHF

Labo 1 - Thermal demand characterization
The heating and ECS demand is derived from SIA 380/1 tables, the building SRE and envelope factors. An hourly profile is built with the MeteoSuisse temperature series and a simplified ECS schedule (08:00-20:00). Thermal inertia and foisonnement factors are applied to approximate diversity between buildings.

Key outputs:
- Annual heating: 17 905.3 MWh/an
- Annual ECS: 8 242.5 MWh/an
- Annual total heat: 26 147.8 MWh/an
- Peak heat demand (hourly): 9 259 kW
- Peak demand with foisonnement: 7 222 kW

Labo 2 - Electricity and PV self-consumption
Electricity loads are read from the consumption workbook. PV production is modelled with PVGIS profiles for east/west orientations and installed surface. Autoconsumption and autonomy are computed on an hourly basis.

Key outputs:
- Annual electricity load: 23 371.9 MWh/an
- Annual PV production: 19 318.9 MWh/an
- Autoconsumption: 76.5%
- Autonomy: 63.3%
- Storage capacity for 100% autoconsumption: 4 052.6 MWh
- Peak load: 5 381 kW
- Peak PV: 5 585 kW

Labo 3 - District heating network optimization (focus)
The optimization uses the road network as the backbone and connects buildings and resources to the nearest street edge. Each pipe segment is assigned a diameter from a discrete catalogue and sized to carry the required mass flow at Delta T = 30C. The objective is to minimize a linearized CAPEX function based on diameter and length. The heat demand is taken from the Labo 1 peak power (after foisonnement).

Key outputs:
- Network length: 14.77 km
- CAPEX (pipes only): 16 736 196 CHF
- Annual heat served: 26 147.8 MWh/an
- Energy density: 1 770.3 MWh/km/an
- Peak heat demand (hourly): 9 259 kW
- Smoothed peak (8h rolling): 8 752 kW

Supply dispatch (optimized):
- rhone: 4 161.1 kW (119 457 kg/h)
- nappe: 2 000.0 kW (57 416 kg/h)
- nappe_ste_marguerite: 713.6 kW (20 485 kg/h)
- cad: 347.6 kW (9 980 kg/h)

Limitations and why the sizing can be too large:
- Peak-based sizing: the network is sized to the hourly peak demand. Short peaks drive large diameters even when the energy-weighted demand is lower. The 8-hour smoothed peak is about 8 752 kW versus 9 259 kW instantaneous.
- No temporal diversity inside the optimization: foisonnement is applied once to building peaks, but the optimization still assumes the full diversified peak occurs simultaneously on every branch.
- No thermal losses or return temperature constraints: heat losses, pump constraints, and operational limits are not modelled, so margins are implicitly absorbed by pipe sizing rather than control strategies.
- Static resource dispatch: resources are constrained only by max power, with no cost hierarchy, startup limits, or seasonal availability.

Suggested next steps:
- Re-size pipes based on a design percentile (e.g., 95th) or use the 8-hour smoothed peak as the reference.
- Add pipe heat losses, pumping power, and velocity constraints to avoid overly large diameters.
- Validate demand with measured profiles and update foisonnement per affectation.
- Evaluate staged deployment to prioritize high energy density corridors.